

CSE 440: Introduction to Artificial Intelligence

Prof. Yu Kong

Outline

- Logistics
- What is AI?
- What is this course?
- Intelligent agents

Team

- **Instructor:** Prof. Yu Kong
- **Office hours:** After class or by appointment through Piazza
- **Teaching Assistants**
 - Wenliang Guo
 - Bingheng Li
 - Karan Choudhary
- **Syllabus:** <https://d2l.msu.edu/d2l/home/2387354>
- **Piazza:** <https://piazza.com/msu/spring2026/cse440>
 - All correspondence must be through Piazza

Week	Date	Tuesday Wells Hall B115 10:20-11:40AM	Thursday Wells Hall B115 10:20-11:40AM	Note
1	Jan 13 & 15	Introduction/Probability	Heuristic Search	
2	Jan 20 & 22	Adversarial Search	Lab	
3	Jan 27 & 29	Constraint Satisfaction Search Problem	Markov Decision Process I	<u>Assg. 1</u>
4	Feb 3 & 5	Markov Decision Process II	Lab	<u>Assg. 2</u>
5	Feb 10 & 12	Reinforcement Learning	Reinforcement Learning	
6	Feb 17 & 19	Reinforcement Learning II	Lab	<u>Assg. 3</u>
7	Feb 24 & 26	Review	Mid-term Exam	
8	Mar 3 & 5	Spring Break	Spring Break	
9	Mar 10 & 12	Bayesian Network I	Bayesian Networks II	
10	Mar 17 & 19	Machine learning: Perceptron and Logistic Regression	Lab	<u>Assg. 4</u>
11	Mar 24 & 26	Machine Learning: Neural Networks	Convolutional Neural Networks	
12	Mar 31 & Apr 2	Convolutional Neural Networks	Deep learning	
13	Apr 7 & 9	Face Recognition	Lab	<u>Assg. 5</u>
14	Apr 14 & 16	Generative AI	Generative AI	
15	Apr 21 & 23	Final Review	Q & A	

Final exam: Friday, 5/01/2026 from 7:45 AM - 9:45 AM in Wells Hall B115

Evaluation

- Attendance (10%)
- Quizzes (10%)
- Homework assignments (20%)
- Lab (25%)
- Midterm exam (15%)
- Final exam (20%)

Timing

- Assignments need to be turned in D2L before the due date. Late assignments will **only** be accepted for exceptional circumstances.
- Make-up for exams will only be granted under exceptional circumstances. Instructor reserves the right to deny requests for make-up.

Attendance

- Attendance at all regularly scheduled class meetings is a requirement of this course. Students with unexcused absences for more than three consecutive class meetings prior to the middle of the term may be dropped from the course for non-attendance.
- We will randomly check attendance throughout the semester.
- **Note: Attendance cheating is certainly an incident of academic dishonesty and it could be subject to the filing of an academic dishonesty report (ADR).**

Academic Integrity

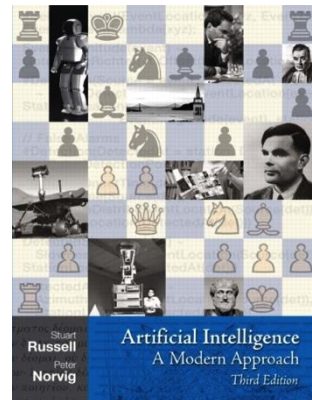
- The Department of Computer Science expects all students to adhere to MSU's policy on Integrity of Scholarship and Grades. Information about MSU policy regarding academic integrity is available at <https://www.msu.edu/~ombud/academic-integrity/index.html>.

Communication

- Piazza for all communications
- Do NOT send emails to instructor or TAs
- HW/Exam submitted via email will NOT be graded
- Install Piazza App on phone, tablet, etc.
- Your responsibility to check regularly

Textbook

- Not a course textbook, and not required, but recommended for students who wants to read more.
- Russell & Norvig, Artificial Intelligence: A Modern Approach



[https://github.com/yanshengjia/ml-road/blob/master/resources/Artificial%20Intelligence%20-%20A%20Modern%20Approach%20\(3rd%20Edition\).pdf](https://github.com/yanshengjia/ml-road/blob/master/resources/Artificial%20Intelligence%20-%20A%20Modern%20Approach%20(3rd%20Edition).pdf)

Outline

- Administrative stuff
- **What is AI?**
- What is this course?
- Intelligent agents

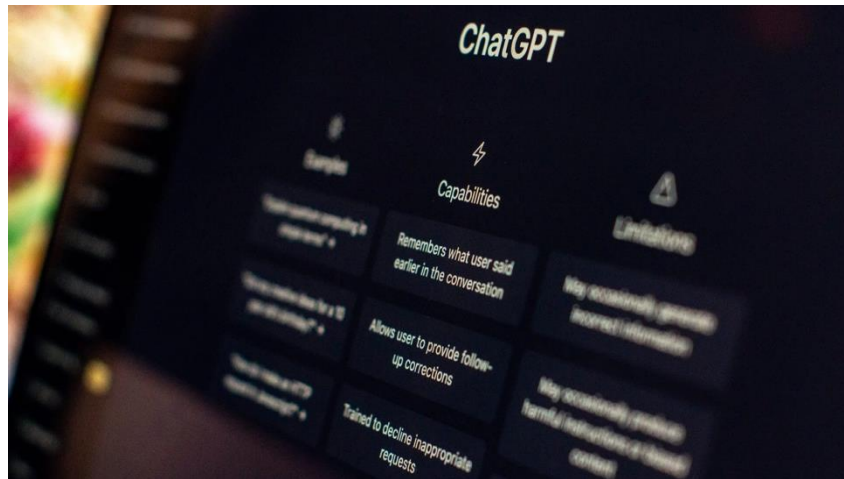
Hollywood AI



News in AI



News in AI



What can AI do?

face recognition



self driving cars



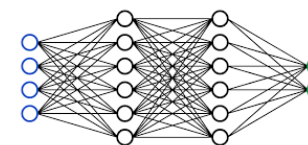
conversation AI



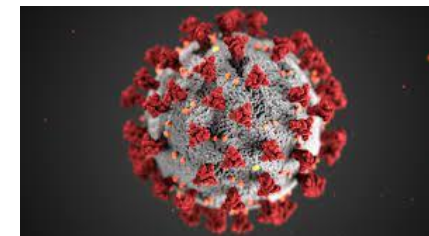
disease diagnosis



Chest CT scan



Deep learning model



COVID-19

Fears

What was Hawking's warning about AI?

Hawking's biggest warning is about the rise of artificial intelligence: It will either be the best thing that's ever happened to us, or it will be the worst thing. If we're not careful, it very well may be the last thing. Oct 16, 2018



Who said AI is a threat to humanity?

Elon Musk

Elon Musk has said he believes AI is “one of the biggest threats” to humanity, and that the UK’s AI Safety Summit was “timely” given the scale of the threat. The billionaire said AI was an “existential risk” because humans for the first time were faced with something “that is going to be far more intelligent than us”. Nov 1, 2023

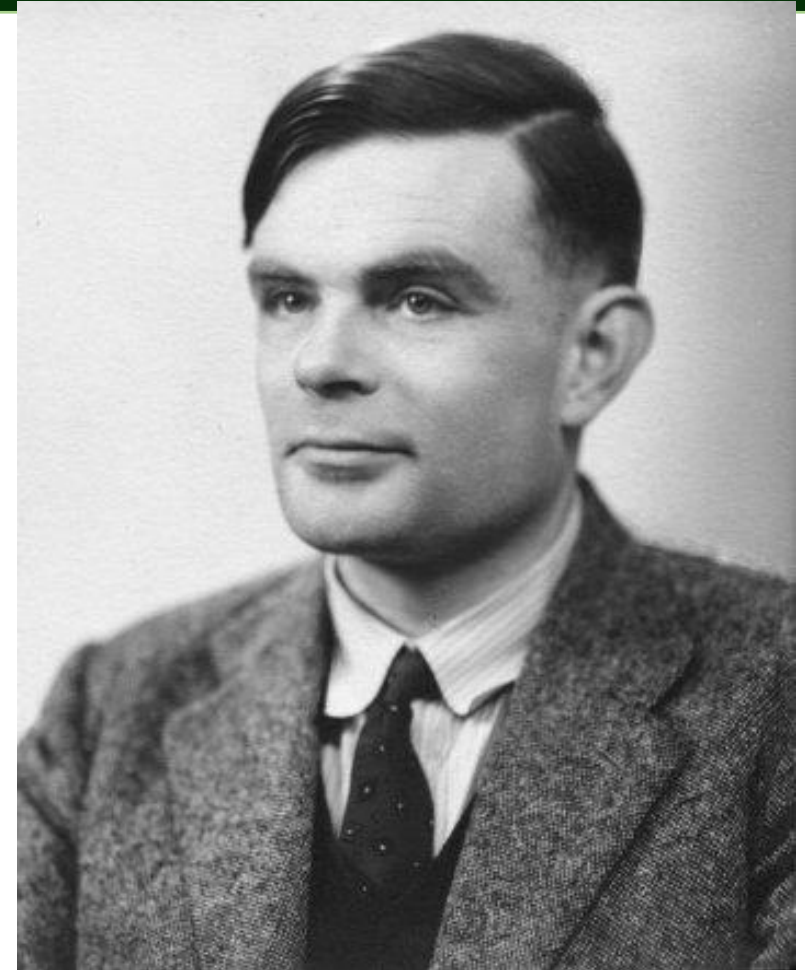


What is AI?

The science of making machines that:

Turing Test

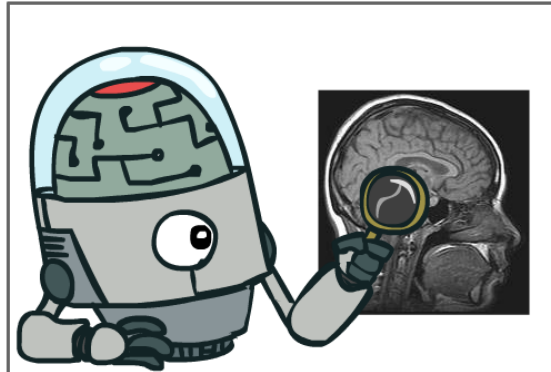
- In 1950, Turing defined a test of whether a machine could “think”
- “A human judge engages in a natural language conversation with one human and one machine, each of which tries to appear human. If judge can’t tell, machine passes the Turing test”



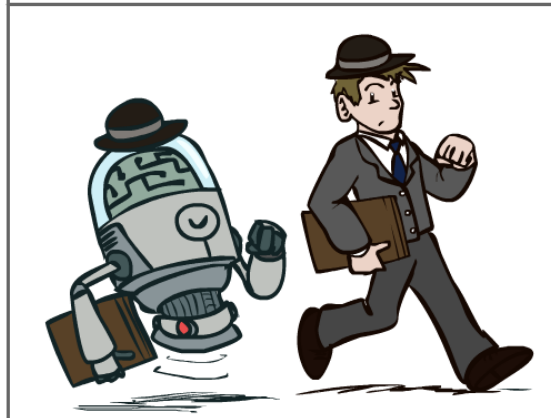
What is AI?

The science of making machines that:

Think like people

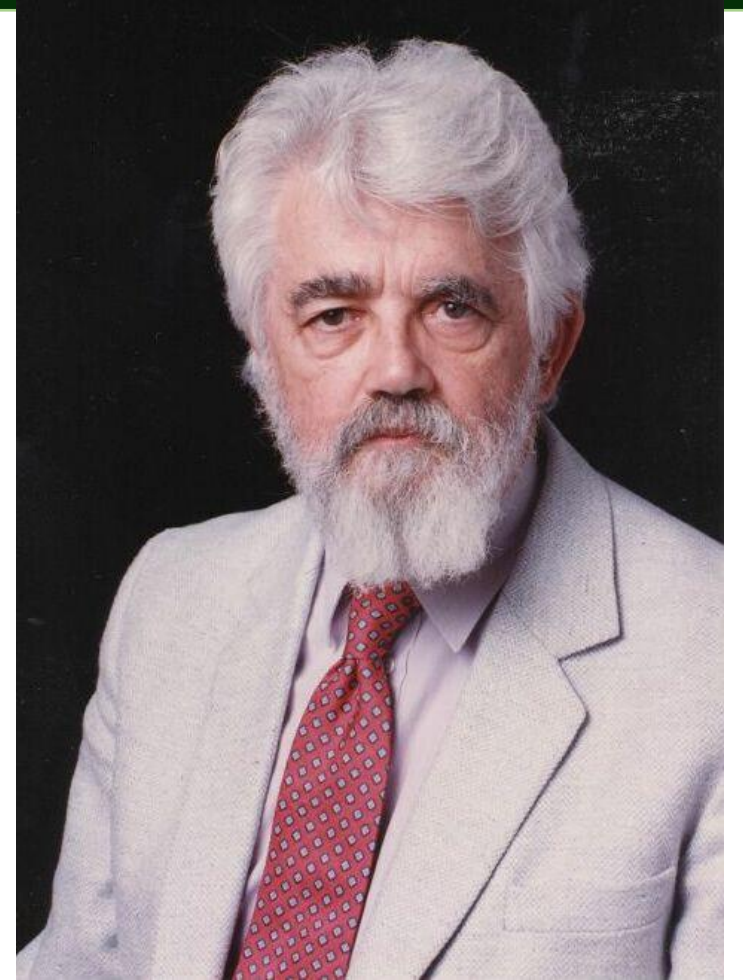


Act like people



AI Definition by John McCarthy

- What is artificial intelligence
 - It is the science and engineering of making intelligent machines, especially intelligent computer programs
- What is intelligence
 - Intelligence is the computational part of the ability to achieve goals in the world



<http://www-formal.stanford.edu/jmc/whatisai/whatisai.html>

Rational Decisions

We'll use the term **rational** in a very specific, technical way:

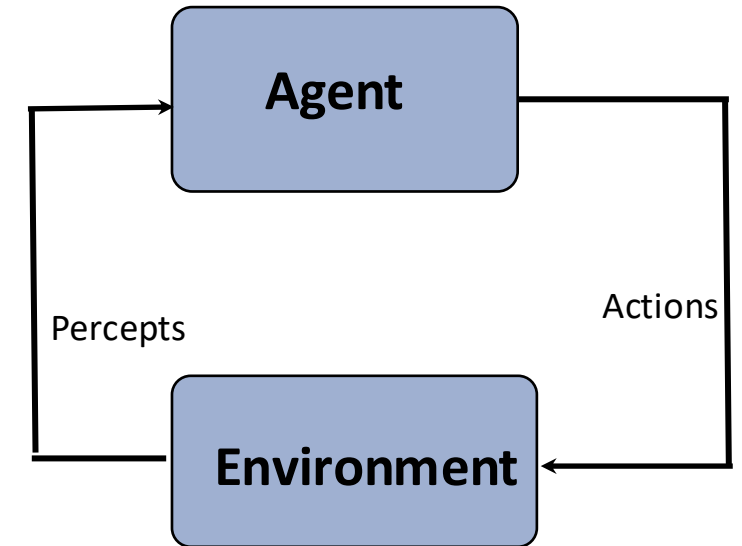
- Rational: maximally achieving pre-defined goals
- Rationality only concerns what decisions are made
(not the thought process behind them)
- Goals are expressed in terms of the **utility** of outcomes
- Being rational means **maximizing your expected utility**

A better title for this course would be:

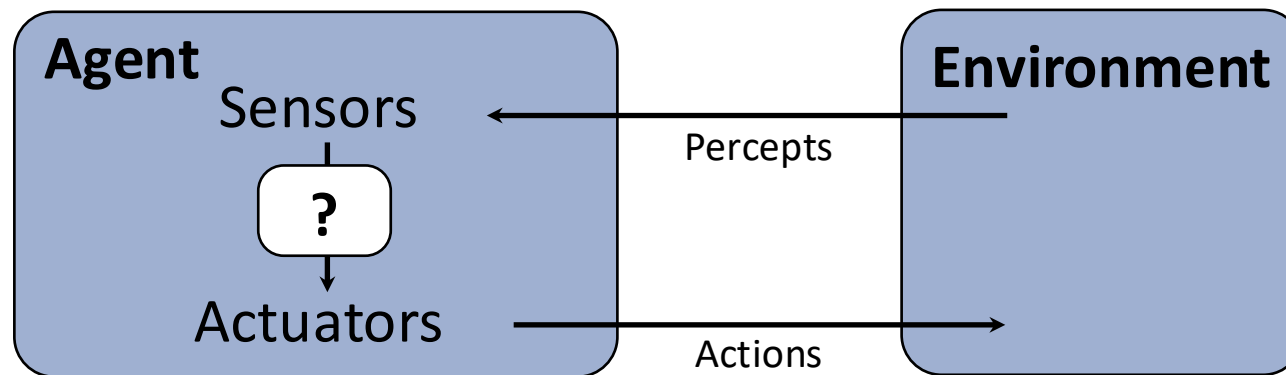
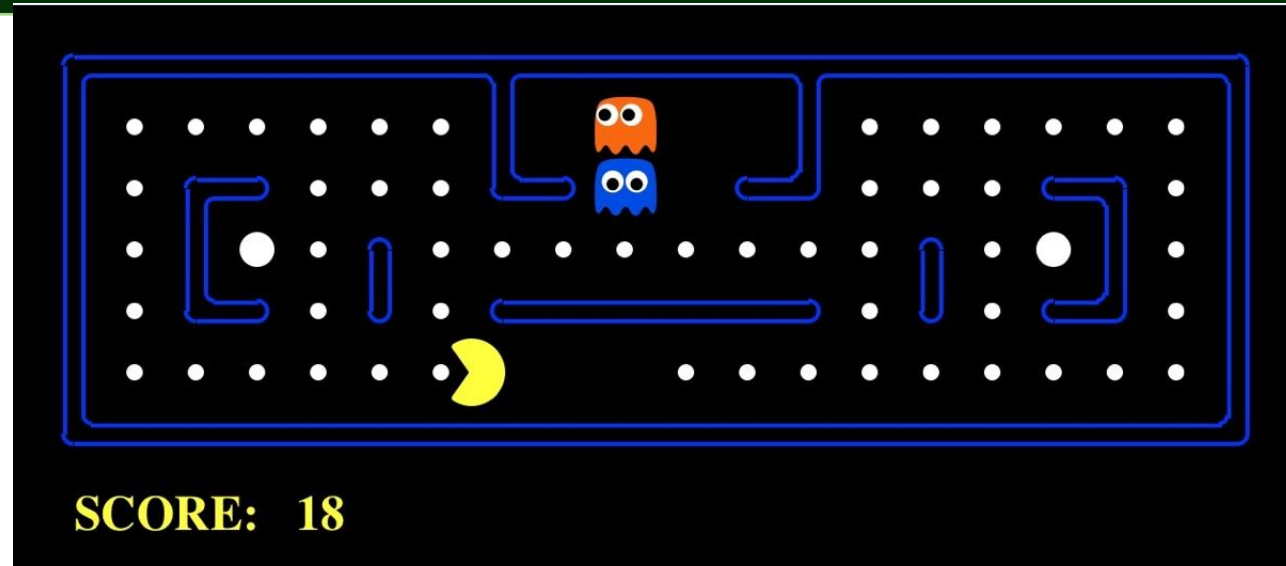
Computational Rationality

Rational Agents

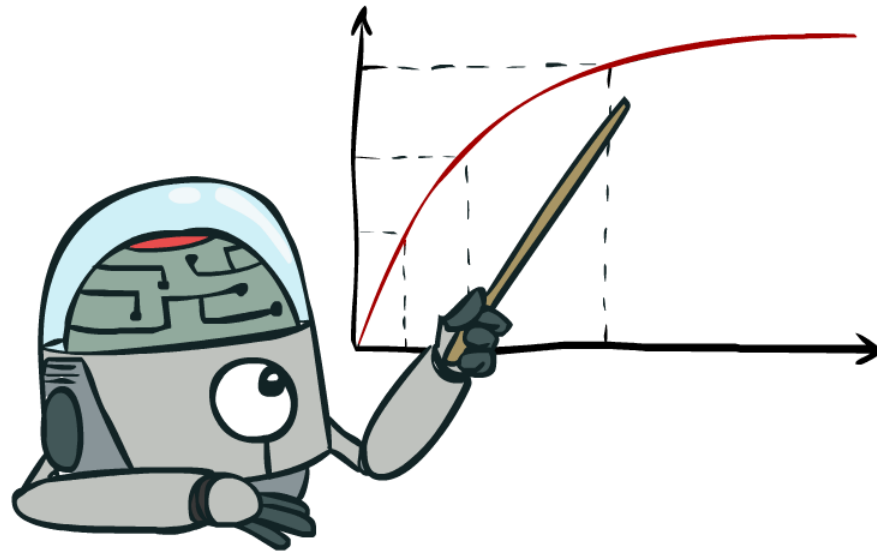
- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
- Characteristics of the **percepts**, **environment**, and **action space** dictate techniques for selecting rational actions



Pac-Man as an Agent

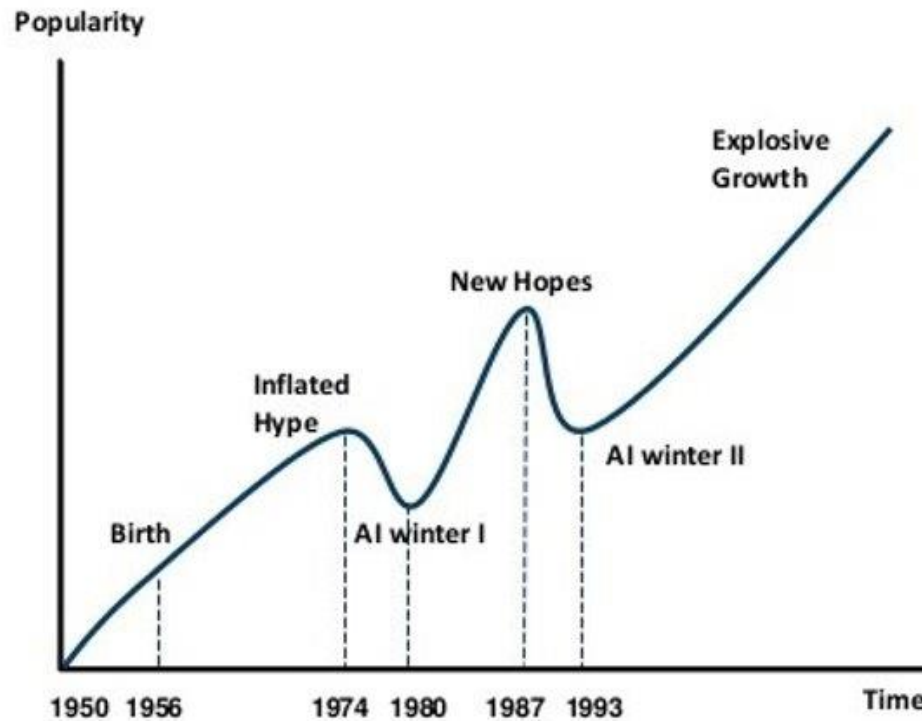


Maximize Your Expected Utility



History of AI - AI Summers and Winters

AI HAS A LONG HISTORY OF BEING “THE NEXT BIG THING” ...



Timeline of AI Development

- **1950s-1960s:** First AI boom - the age of reasoning, prototype AI developed
- **1970s:** AI winter I
- **1980s-1990s:** Second AI boom: the age of Knowledge representation (appearance of expert systems capable of reproducing human decision-making)
- **1990s:** AI winter II
- **1997:** Deep Blue beats Gary Kasparov
- **2006:** University of Toronto develops Deep Learning
- **2011:** IBM's Watson won Jeopardy
- **2016:** Go software based on Deep Learning beats world's champions

<https://www.actuaries.digital/2018/09/05/history-of-ai-winters/>

<https://searchenterpriseai.techtarget.com/infographic/Timeline-of-AI-winters-casts-a-shadow-over-todays-applications>

History of AI - First AI Boom (1956-1974)

- 1950s: The concept of AI is proposed
- 1960s: Great amount of money invested by DARPA (the US Defense Advanced Research Project Agency)
- 1958: The 1st geometry theorem proving machine
- 1964: The 1st algebra word problem solver: STUDENT
- 1966: The 1st chatbot: ELIZA
- ...

History of AI - First AI Winter (1974-1980)

- 1969: Marvin Minsky's devastating criticism of perceptrons
- 1974: Lighthill report states “In no part of the field (of AI) have the discoveries made so far produced the major impact that was then promised”
- 1974-1980: DARPA cuts its funding for the field AI significantly



History of AI - Second AI Boom (1980-1987)

- 1980: The rise of expert system
- 1982: The proposal of “Hopfield net” and the promotion of “backpropagation”
- 1980-1987: More money for AI research invested by various countries, such as Japan, US, UK

History of AI - Second AI Winter (1987-1994)

- 1987: A collapse of the market for specialized AI hardware
- 1987: DARPA again cuts funding for AI research
- Late 1980s: Expert systems are proven too expensive to maintain
- 1991: Japanese researchers fail to deliver on goals of holding conversations, interpreting images and achieving humanlike reasoning

History of AI - Third AI Boom (1994-present)

- 1997: Deep Blue beats Gary Kasparov
- 2006: The birth of deep learning
- 2011: IBM's Watson beats Jeopardy
- 2016: AlphaGo wins world's champions
- Breakthrough in computation power, availability of big data, more and more real-world applications, ...



What can AI do now?

- Q: Which of the following can be done at present?
- Win against any human at chess?
- Win against the best humans at Go?
- Play a decent game of tennis?
- Unload any dishwasher in any home?
- Drive safely along the highway?
- Drive safely along streets of Chicago?
- Buy a week's worth of groceries on the web?
- Buy a week's worth of groceries at Meijer's?
- Discover and prove a new mathematical theorem?
- Perform a surgical operation?
- Translate spoken Chinese into spoken English in real time?
- Win an art competition?
- Write an intentionally funny story?
- Construct a building?



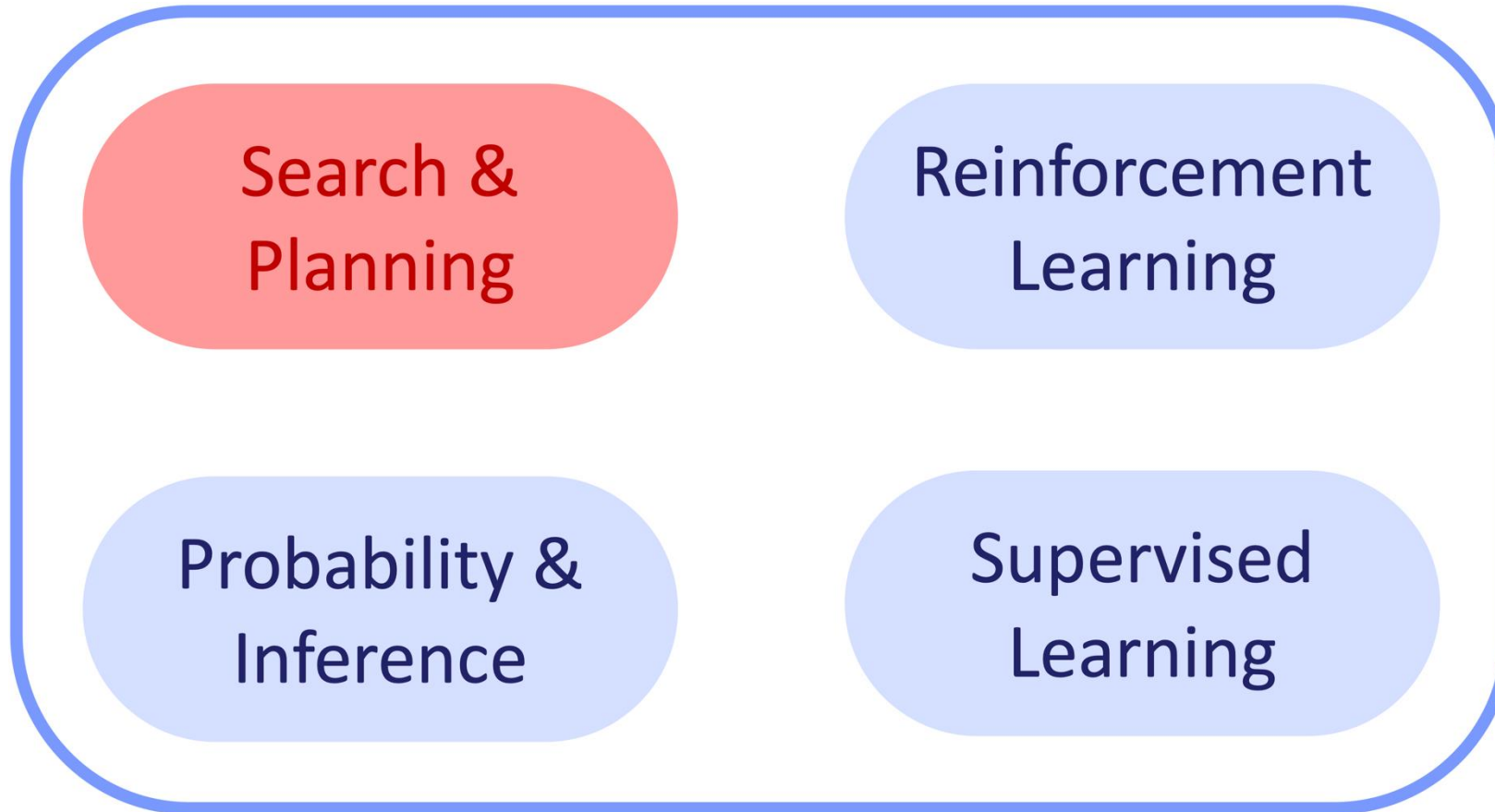
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Course Topics

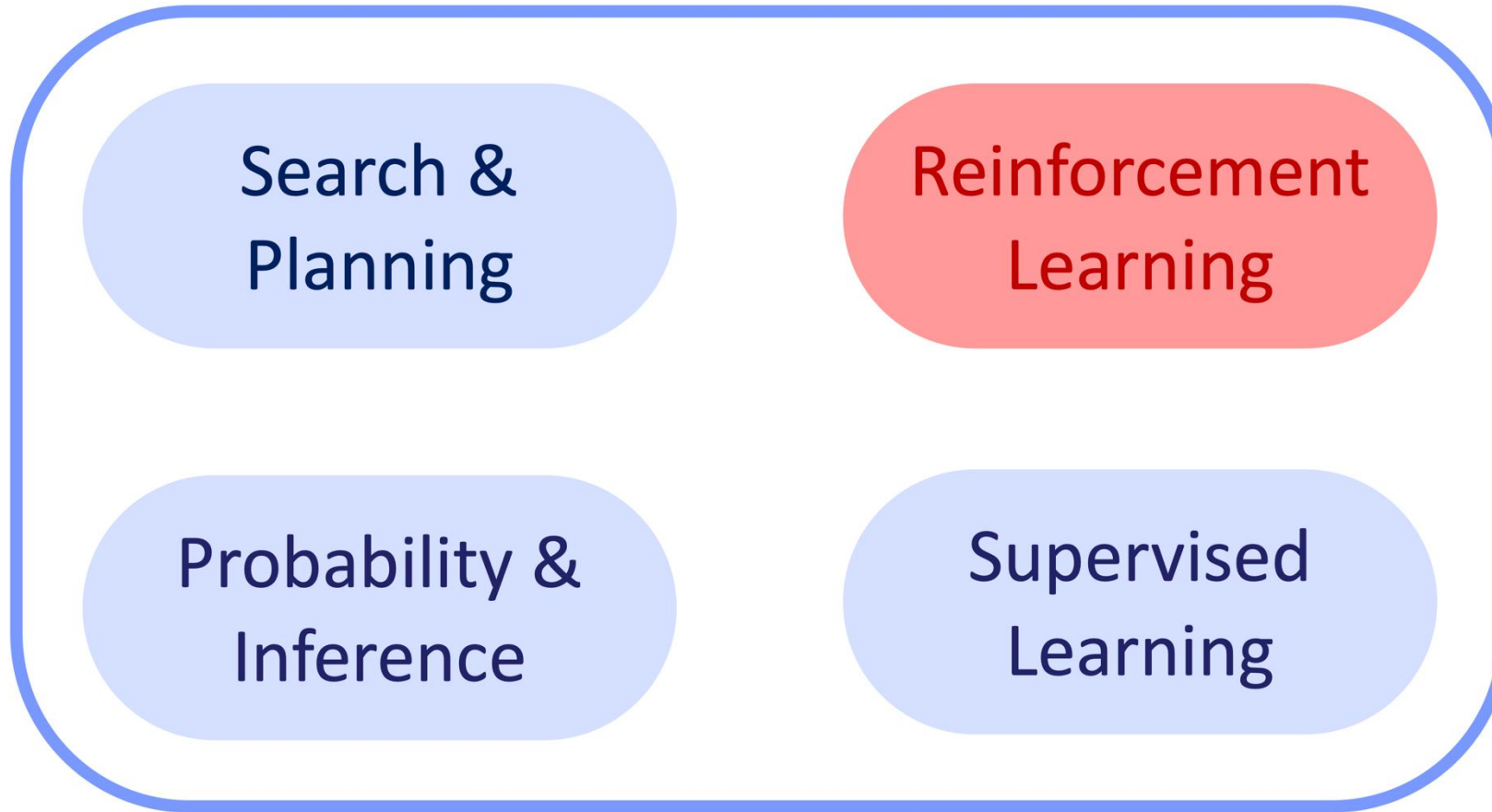
- What will AI do?
- What are the basics of how AI works?
 - Intelligent agents, e.g. search problem, adversarial search
 - Logic-based inference, e.g. constraint satisfaction
 - Decision Making
 - Reinforcement Learning
 - Machine learning, e.g. Neural Networks

Schedule



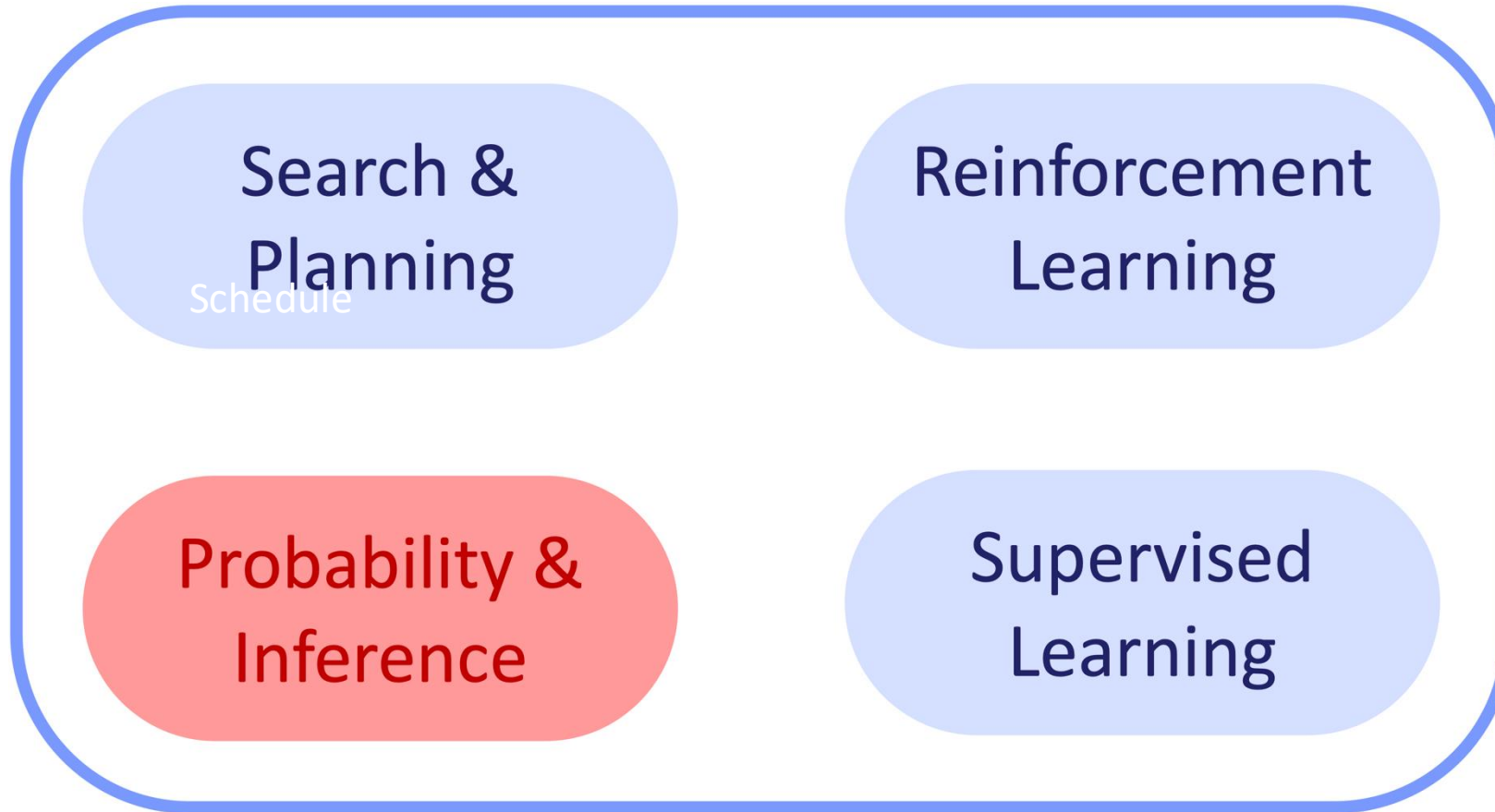
How can I find a *sequence of best decisions* for a *particular* situation?

Schedule



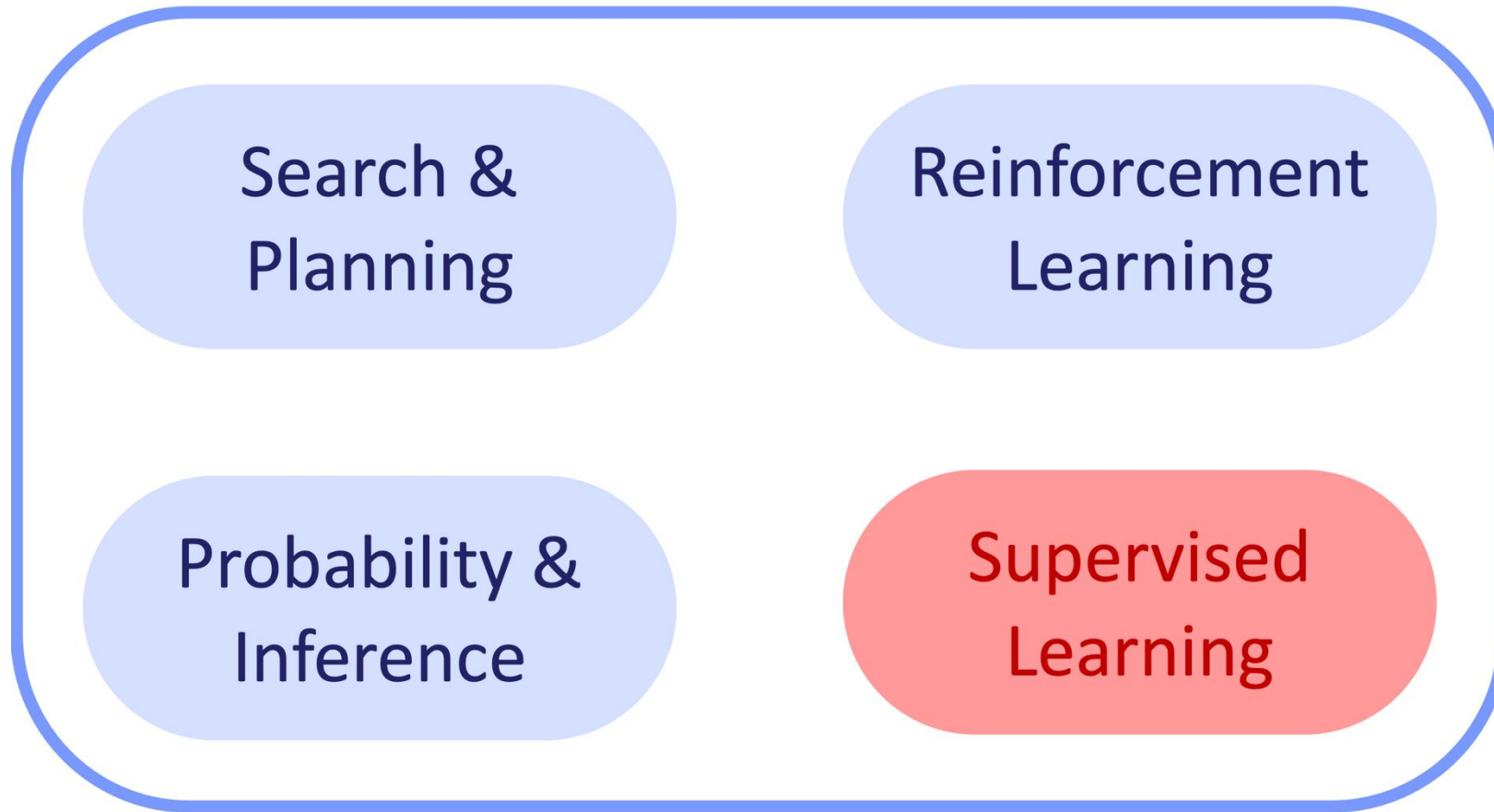
How can I find **rules (policy)** to make best decisions for **any** situation?

Schedule



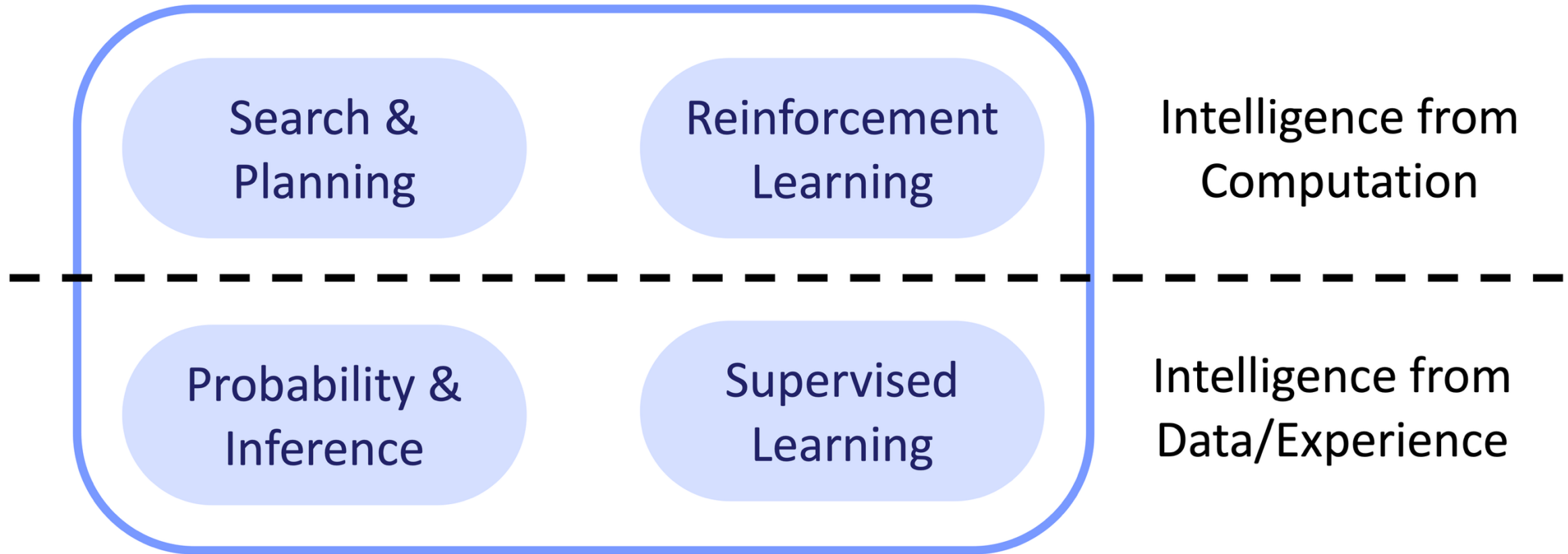
How can I make sense of *uncertainty* in the world?

Schedule

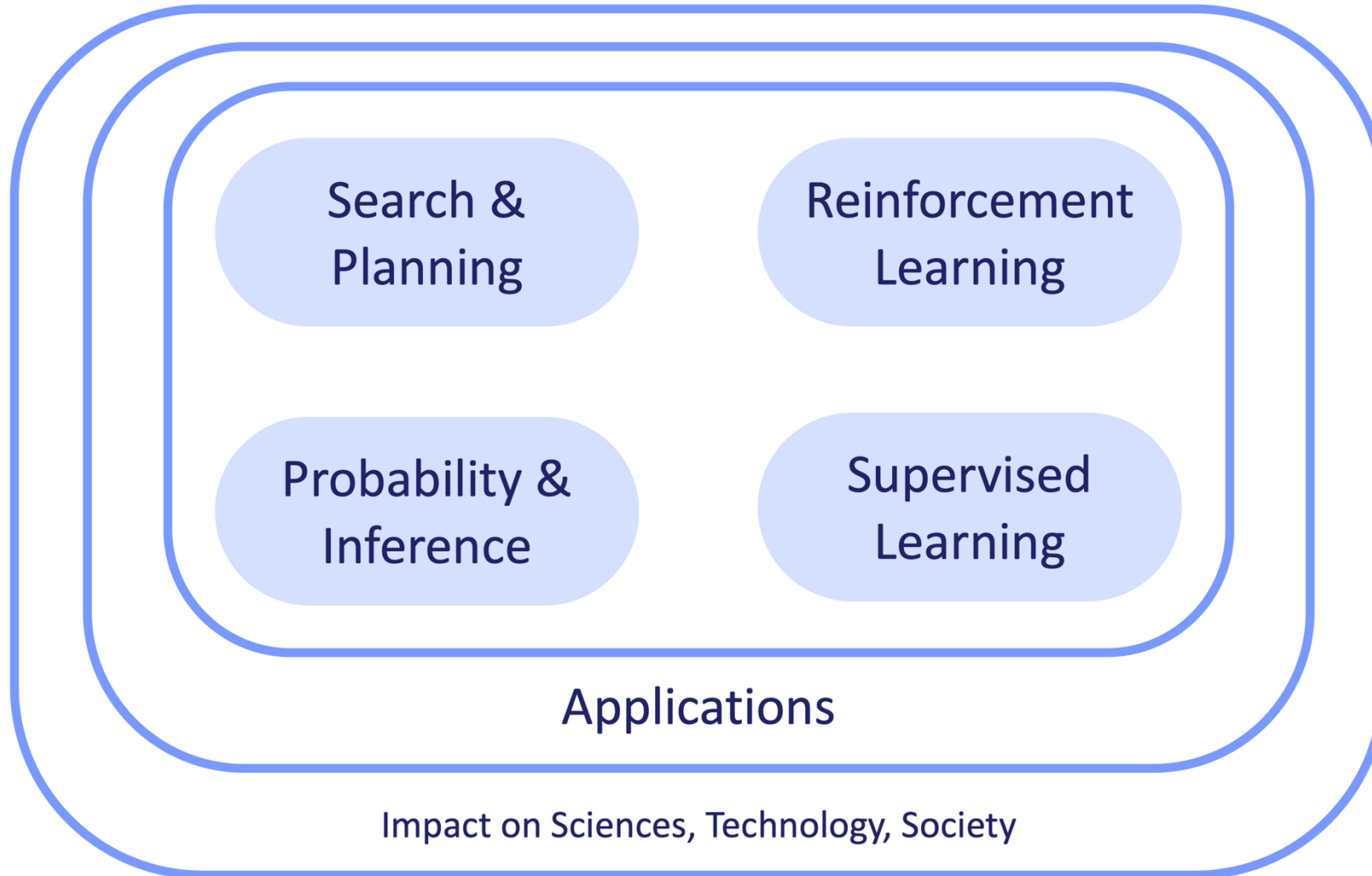


How can I learn a *model* of the world from *data*?

Schedule



Schedule



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Reflex Agents VS Planning Agents

- Reflex agents:
 - Choose action based on current state(and maybe memory)
 - May have memory or a model of the world's current state
 - Do not consider the future consequences of their actions
 - Consider how the world **IS**
- Can a reflex agent be rational?

Reflex Agents VS Planning Agents

- Planning agents:
 - Decisions based on estimated consequences of actions
 - Must have a model of how the world evolves in response to actions
 - Must formulate a goal (test)
 - Consider how the world **WOULD BE** as opposed to how the world **IS**